

Logic Block Form Instructions

MC12-500

**Implementation
Hiway Gateway - 2**

Logic Block Form Instructions

**MC12-500
9/95**

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About This Publication

This publication provides a guide for completing the Logic Block Configuration Form MC88-501 and the Logic Block Comments Entry Form MC88-502.

This publication supports Release 500.

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INTRODUCTION

Section 1

1.1 PURPOSE

This publication provides general and specific instructions for completing the Logic Block Configuration Form and the Logic Block Configuration Comments Form. The instructions include a sample Logic Block Configuration Form and a sample Comments Form. The Logic Block Configuration Form is designed to facilitate collection and organization of the data required to define logic blocks, which are functional entities executed in Multifunction Controllers. The logic block definitions (up to 128 for each Multifunction Controller) are part of the database that is configured for, and down-line loaded to, Multifunction Controllers (Figure 1-1).

1.2 SCOPE

This publication describes the method of defining logic blocks on 8 1/2 x 11 inch paper forms that facilitate later entry into the TDC 3000^X system. It also describes the method of entering comments regarding logic block use on paper forms that facilitate entry into the system.

1.3 PREREQUISITES

Before using this publication for filling in configuration and comments forms, the user should

- understand the fundamentals of TDC 3000^X System configuration, particularly as they apply to the Multifunction Controller, and specifically to logic blocks. (Refer to the *Configuration Data Collection Guide*).
- determine, for a given Multifunction Controller, all process-related and special logic requirements, and record these requirements in lists and/or appropriate diagrams.

1.4 RELATED DOCUMENTS

The following publications should be available for reference while using this publication:

Title	Publication No.	Binder
Configuration Data Collection Guide	SW12-500	Implementation/Startup & Reconfiguration - 2
Logic Block Data Entry	MC11-500	Implementation/Hiway Gateway - 2

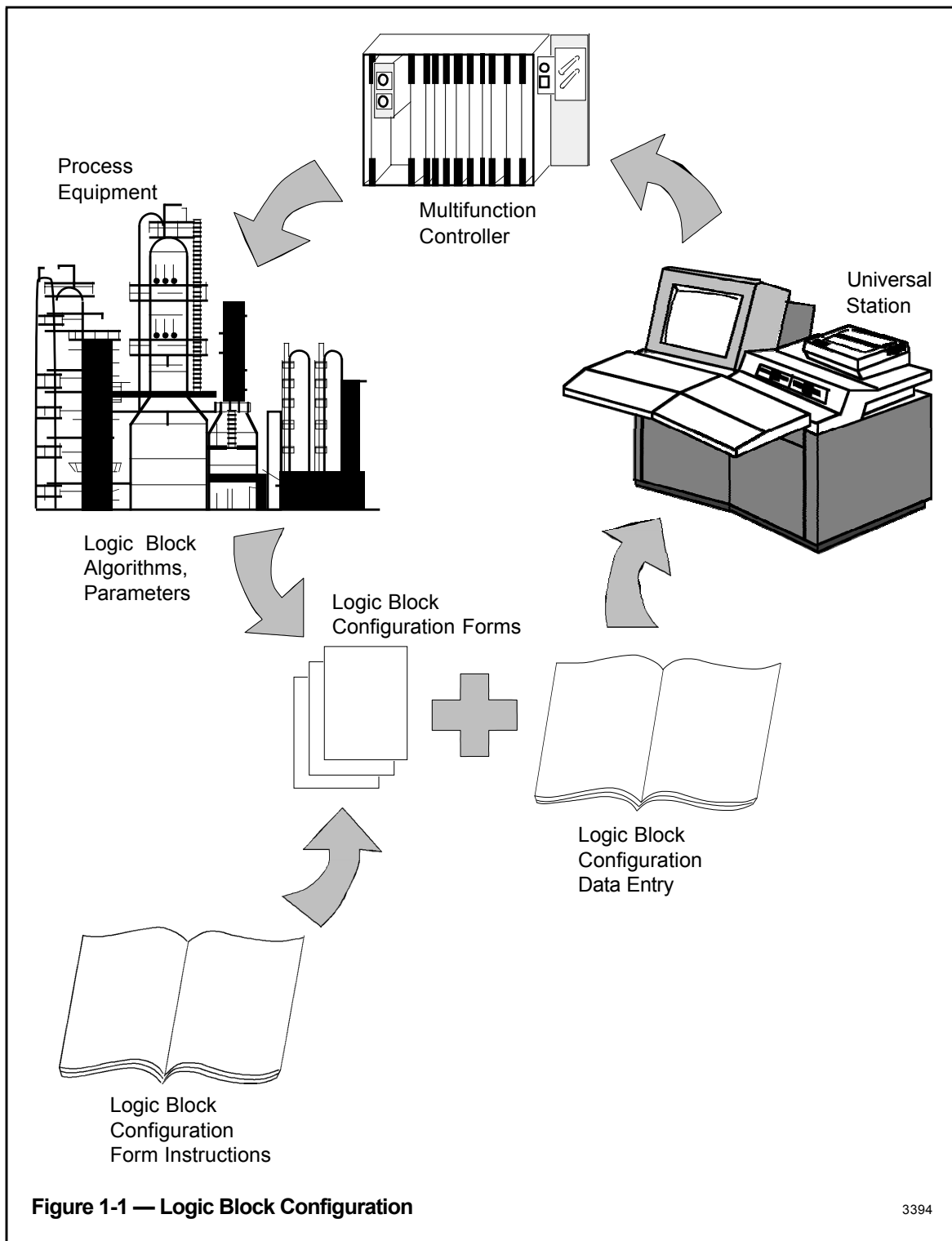


Figure 1-1 — Logic Block Configuration

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OVERVIEW

Section 2

A Logic Block is similar in function to a logic element that can be implemented in dedicated hardware such as binary electronic circuitry. Logic Blocks are actually firmware entities in the Multifunction Controller that act on details of logic definition (algorithms, inputs, and outputs) that are entered into RAM by Logic Block Configuration Data Entry.

Each Multifunction Controller can process up-to-128 logic blocks. Any of the 128 logic blocks can be configured to execute any one of 12 algorithms. The 12 algorithms are described in Section 3, Table 3-3. They are listed by logic block type, which is determined by the number of inputs and outputs that can be configured for a given logic block.

The Engineer who is responsible for configuring logic blocks in a given Multifunction Controller should proceed generally as follows:

1. Determine the logic function requirements of the Multifunction Controller to be configured. Considerations should include interactions between the Multifunction Controller in question and other system components, particularly other Multifunction Controllers on the same C-Link, and interactions between logic-block functions and functions included in all programs that are intended for use in the Multifunction Controller.
2. Describe and record logic-block use in generic terms, using traditional methods such as logic diagrams, sequence diagrams, etc., that are familiar to the engineer who is doing the logic design.
3. For each of the (up-to-128) logic blocks, translate details of each logic element into logic conventions used for logic-block definition, and enter this data on the Logic Block Configuration Forms supplied with the system.
4. On the Logic Block Comments Sheet(s) provided with the system, describe functions of groups of interconnected logic blocks, and the relationships between functional groups of logic blocks. Because logic blocks are entered on the forms and into the system as single entities in which the intended use of related blocks may not be clear, it is important that a record of design intent is made at the time of original design. This is particularly important when design changes must be made significantly later than the original design, and by personnel other than the original designers.

NOTE

The comments should be recorded on the sheets provided, because these sheets facilitate entry into the system (through the Text File Editor function) for convenient retrieval.

Graphic representations of four types of logic blocks, completed configuration and comment forms built from this data, and an associated Text Builder display are depicted in Figure 2-1.

SPECIFIC INSTRUCTIONS

Section 3

3.1 LOGIC BLOCK CONFIGURATION FORM

The Logic Block Configuration Form, shown on the following page, is printed on an 8 1/2 by 11 inch sheet. Each page is formatted to collect all the data required to define four logic blocks. The form is divided into three parts, in the following manner:

Lines 0.0 thru 0.3	Pathname—provides data entry areas for the system identifiers with which the logic block is associated (device, volume, and filename). This data is not actually part of the data required to define logic blocks; its primary function is to enable completed forms to be grouped for later entry of data into the system.
Lines 1.0 thru 5.0	Configuration Lines—in which the word "lines" refers to data entry areas on the form and that correspond to the Logic Block Configurator - Text Builder display (refer to the <i>Logic Block Data Entry</i> publication for additional details). Each set of lines 1.0 through 5.0 defines a single logic block.
Logic Block Types	Graphic representations of the five different types of logic blocks, including available and/or required inputs and outputs, and a list of the algorithms included in each type.

Instructions for filling out the Logic Block Configuration Form are presented in Table 3-1. The instructions in this table are sequenced in the same order as they are presented on the form and on the Logic Block Configurator - Text Builder display, and each instruction is referenced by a line number that appears on the form.

Definitions of the entries required for the algorithm and its inputs and outputs are listed in Tables 3-2 through 3-4.

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LOGIC BLOCK CONFIGURATION FORM

LBC - TEXT BUILDER DISPLAY

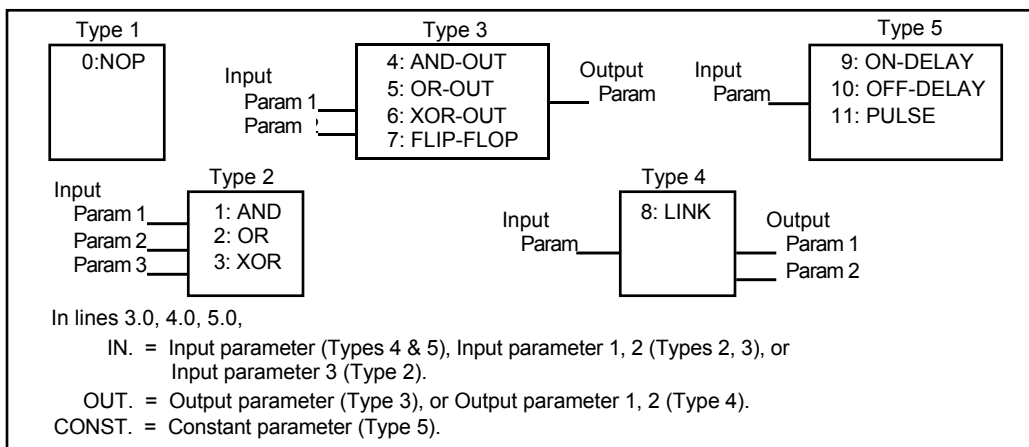
0.0 Pathname

0.1 Device | N | E | T | _ | _ | _ | _ | _ | _ | _ |

0.2 Volume | V | 1 | 0 | 0 | _ | _ | _ | _ | _ | _ |

0.3 Filename | L | B | 0 | 1 | 1 | 4 | 1 | 0 | _ | _ |

1.0 BLOCK NUMBER	1 0 2
2.0 ALGORITHM	0 3 / K O R _ _ _ _ _ _
3.0 IN.	L B (1 0 0) _ _ _ _ _ _ _
4.0 IN./OUT./CONST.	+ - + - 0 +) _ _ _ _ _ _ _
5.0 IN./OUT.	_ _ _ _ _ _ _ _ _ _ _ _
1.0 BLOCK NUMBER	1 0 3
2.0 ALGORITHM	1 1 / P U L S E _ _ _ _
3.0 IN.	C I (1 0 1) . P R E D E V F L _ _
4.0 IN./OUT./CONST.	+ - + - 0 +) _ _ _ _ _ _ _
5.0 IN./OUT.	_ _ _ _ _ _ _ _ _ _ _ _
1.0 BLOCK NUMBER	1 0 4
2.0 ALGORITHM	0 7 / F L I P - F L O P
3.0 IN.	L B (1 0 2) _ _ _ _ _ _ _
4.0 IN./OUT./CONST.	+ - + - 0 +) _ _ _ _ _ _ _
5.0 IN./OUT.	_ _ _ _ _ _ _ _ _ _ _ _
1.0 BLOCK NUMBER	1 0 5
2.0 ALGORITHM	0 8 / L I N K _ _ _ _
3.0 IN.	L B (1 0 4) _ _ _ _ _ _ _
4.0 IN./OUT./CONST.	+ - + - 0 +) _ _ _ _ _ _ _
5.0 IN./OUT.	+ - + - 2 8) _ _ _ _ _ _ _



REFERENCES: Logic Block Form Instructions in the Implementation/Hiway Gateway - 1 binder
 Logic Block Data Entry in the Implementation/Hiway Gateway - 1 binder

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Table 3-1 — Entries for Logic Block Configuration Form

LINE	NAME	DATA ENTRY	REMARKS
LBC - TEXT BUILDER DISPLAY			
0.0	Pathname	\$Fn (floppy drive n)	Enter User Pathname
0.2	Volume	XXXX (user-defined in Network configuration)	
0.3	Filename	LBxxyyzz, where xx = Hiway Number (01-20) yy = Box Number (01-31) zz = Serial Number (00-99)	Configured in Network. Box Number of Multifunction Controller. User number for revision level of this filename.
1.0	BLOCK NUMBER	Enter three digits, from 001 through 128.	Block Number must be unique within a given Multifunction Controller. Logic blocks that are inputs must have a block number lower than that of the blocks for which they are inputs.
2.0	ALGORITHM	Enter one or two digits (0 through 11) that correspond to the desired algorithm or enter algorithm name.	Algorithm numbers/names are shown on forms. Numbers, names and descriptions are in Table 3-2.
3.0	IN.	Enter up-to-18 characters to select the desired input parameter.	In Line 3.0, IN. = Input parameter (number unspecified in Types 4 and 5; 1 in Types 2 and 3. Allowable input parameters are in Table 3-3.
4.0	IN./OUT./CONST.	Enter: - up-to-18 characters to select the desired input parameter (IN.), or - up-to-17 characters to select the desired output parameter, (OUT.), or - an integer number between 0001 and 9999 to specify a time constant (CONST.).	In Line 4.0, IN. = Input parameter 2 (Types 2 and 3); OUT. = Output parameter (Type 3), or Output parameter 1 (Type 4); CONST. = Constant parameter (Type 5). Allowable parameters for - Inputs are in Table 3-3 - Outputs are in Table 3-4 Constant parameter specifies time in seconds.
5.0	IN./OUT.	Enter: - up-to-18 characters to select the desired input parameter (IN.), or - up-to-17 characters to select the desired output parameter (OUT.).	In Line 5.0, IN. = Input parameter 3 (Type 2); OUT = Output parameter (unspecified in Type 3, Output parameter 2 in Type 4). Allowable parameters for - Inputs are in Table 3-3 - Outputs are in Table 3-4

Table 3-2 — Logic Block Algorithms

ALGORITHM		FUNCTION	COMMENTS
NO.	NAME		
Type 1 - No Input/No Output			
0	NOP	No operation	No operation, but NOP blocks have logic block numbers like other logic blocks.
Type 2 - Three Input/No Output			
1	AND	Logical AND	All configured inputs must be active (1-state) to produce active state of the block. At least two inputs must be configured. Block state is available for test by programs or as input to another logic block, but is not usable for direct control action (as for interlocks).
2	OR	Logical OR	Any configured input must be active (1-state) to produce active (1-state) block state. At least one input must be configured. Block state is available for test by programs or as input to another logic block, but isn't usable for direct control action (e.g. interlocks).
3	XOR	Exclusive OR	Only one of configured inputs may be active (1-state) to produce active block state. At least two inputs must be configured. Block state is available for test by programs or as input to another logic block, but is not directly usable for control (as for interlocks).
Type 3 - Two Input/One Output			
4	AND-OUT	Logical AND	Comments for AND algorithm (above) apply, except that block state is available through the specified output parameter for control action.
5	OR-OUT	Logical OR	Comments for OR algorithm (above) apply, except that block state is available through the specified parameter for control action.
6	XOR-OUT	Logical OR	Comments for XOR algorithm (above) apply, except that block state is available through the specified output for control action.
7	FLIP-FLOP	Binary storage element	Input parameter 1 active (1-state) sets the flip-flop logic block and element specified output parameter to active state (1-state). Input parameter 2 active resets to 0-state.
Type 4 - One Input/Two Output			
8	LINK	Signal distribution	Input parameter active (1-state) causes both output parameters to be active.
Type 5 - One Input/Constant			
9	ON-DELAY	Delay	Input parameter going active causes block state to go active after the delay time (seconds) specified by the Constant parameter. The block state is available for test by programs or as an input to another logic block, but is not directly available for control action.
10	OFF-DELAY	Delay	Input parameter going active causes block state to go inactive after the delay time (seconds) specified by the Constant parameter. Block state is available to programs or as an input to another logic block, but is not directly available for control action.
11	PULSE	Pulse	Input going active causes block to be active for the time (seconds) specified by the Constant parameter. Block state is available for test by programs and as input to another logic block, but is not directly usable for control action.

Table 3-3 — Input Parameters

POINT TYPE	FORMAT	PARAMETER	VALUE	DESCRIPTION
Modulating Control	LP(xx).zz	xx : slot number zz : variable code	01-16 PTINAL ALHIFL ALLOFL SOA SOB LM OMAN OAUTO OCAS OCOM INITMAN PMAN PAUTO PCAS PCOM PROG	Alarm State Hi alarm Lo alarm DO 1 (LO) DO 2 (HI) Loop-module manual mode Manual mode Auto mode Cascade mode Computer mode Initialization manual mode Program manual mode Program auto mode Program cascade mode Program computer mode Program attribute
Digital input	DI(xxyy).zz	xx : slot number yy : subslot number zz : variable code	01-16 01-16 PV PTINAL	Digital input state Alarm state
Digital output	DO(xxy).zz	xx : slot number y : subslot number zz : variable code	01-16 1-8 OP	Digital output state
Counter input	CI(xxy).zz	xx : slot number y : subslot number zz : variable code	01-16 1-4 STATE RESET STRTSTOP PRESETFL PREDEVFL	Counter active state same as STATE same as STATE Alarm state Pre-alarm state
Analog input	AI(xx).zz	xx : AI point number zz : variable code	01-16 STATE RESET STRTSTOP PTINAL ALHIFL ALLOFL	Accumulation active state same as STATE same as STATE Alarm state HI alarm LO alarm
Timer	TM(xx).zz	xx : timer number zz : variable code	01-32 STATE RESET STRTSTOP TIMOUTFL	Timer active state same as STATE same as STATE Time-up alarm
Flag	FL(xxx)	xxx : flag number	001-256	
Logic block	LB(xxx)	xxx : LB number	001-128	
Others		Time Out Gate Soft Failure Partial Failure	TOG SFTFL PRTFL	

Note: The Boolean complement of any Input Parameter is specified by placing a minus sign (-) before the Input Parameter Code. For example, the code -LP(01).PTINAL, when true, means that Loop 01 is **not** in alarm.

Table 3-4 — Output Parameters

POINT TYPE	FORMAT	PARAMETER	VALUE	DESCRIPTION
Modulating Control	LP(xx).zz	xx : slot number zz : variable code	01-16 SOA SOB OMAN OAUTO OCAS OCOM PMAN PAUTO PCAS PCOM	DO 1 (LO) request DO 2 (HI) request Manual mode request Auto mode request Cascade mode request Computer mode request Program manual-mode request Program auto-mode request Program cascade-mode request Program computer-mode request
Digital Output	DO(xxy).zz	xx : slot number y : subslot number zz : variable code	01-16 1-8 OP	Digital output request
Counter Input	CI(xxy).zz	xx : slot number y : subslot number zz : variable code	01-16 1-4 RESET STRTSTOP	Start/reset request Restart/stop request
Analog Input	AI(xx).zz	xx : AI point number zz : variable code	01-16 RESET STRTSTOP	Start/reset request Restart/stop request
Timer	TM(xx).zz	xx : timer number zz : variable code	01-32 RESET STRTSTOP	Reset/stop request Start/stop request
Flag	FL(xxx)	xxx : flag number	001-256	

Note: A continuous output is specified by placing an asterisk (*) before the Output Parameter Code, and a momentary output is specified by not including the asterisk. For example, *FL(001) specifies continuous output to Flag 001, while FL(001) specifies a momentary output.

3.2 LOGIC BLOCK COMMENTS ENTRY FORM

The Logic Block Comments Entry Form, shown on the following page, is also printed on an 8 1/2 by 11 inch sheet. The form is divided into two parts, in the following manner:

- | | |
|-----------------------------|---|
| Lines
0.0
thru
0.3 | Pathname—provides data entry areas for the system identifiers (device, volume, and filename) with which the set of logic blocks (up-to-128 for a given Multifunction Controller) is associated. Pathname data is not actually part of logic-block comments that are entered into the system; it is intended only to identify each Comments form with a particular Multifunction Controller and data-base file. |
| 1.0 | <p>Comment Lines—in which user comments relating to logic blocks for a given Multifunction Controller are recorded. After logic-block configuration data has been entered into the system (through the Logic Block Configurator function), these comments can be entered into the system through the Text File Editor function. The comments are available to the user for reference</p> <ul style="list-style-type: none"> • on the forms on which they are entered. • on the Text File Editor display. • on text listings (printouts) obtained through the Text File Editor. |

Comments are not executed in the system and, therefore, have no effect on system operation. Comment entries are optional; they can include any information that is meaningful to the user. Comments are very important when logic-block configuration must be correlated with system functions (such as programming) and/or must be modified as a result of revisions to process requirements. Clear and comprehensive comments are particularly important when logic-block modifications and correlation with other system functions must be done by personnel other than those who initially do logic block configuration.

Instructions for filling out the Logic Block Comments Entry Form are listed in Table 3-5.

LBC - TEXT FILE EDITOR DISPLAY

```

0.0 Pathname
0.1 Device      |_|_|_|
0.2 Volume      |_|_|_|_|
0.3 Filename    |_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|_|

```

Number of Logic Block | _ | _ | _ |

[illegible]

Number of Logic Block	1	2	3	4
-----------------------	---	---	---	---

Name of Logo Block		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100		

Number of Logic Block | | |

[illegible]

Number of Logic Block | _ | _ | _ |

Name of the Organization		Date		Page	
1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30
31	32	33	34	35	36
37	38	39	40	41	42
43	44	45	46	47	48
49	50	51	52	53	54
55	56	57	58	59	60
61	62	63	64	65	66
67	68	69	70	71	72
73	74	75	76	77	78
79	80	81	82	83	84
85	86	87	88	89	90
91	92	93	94	95	96
97	98	99	100	101	102
103	104	105	106	107	108
109	110	111	112	113	114
115	116	117	118	119	120
121	122	123	124	125	126
127	128	129	130	131	132
133	134	135	136	137	138
139	140	141	142	143	144
145	146	147	148	149	150
151	152	153	154	155	156
157	158	159	160	161	162
163	164	165	166	167	168
169	170	171	172	173	174
175	176	177	178	179	180
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193	194	195	196	197	198
199	200	201	202	203	204
205	206	207	208	209	210
211	212	213	214	215	216
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223	224	225	226	227	228
229	230	231	232	233	234
235	236	237	238	239	240
241	242	243	244	245	246
247	248	249	250	251	252
253	254	255	256	257	258
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271	272	273	274	275	276
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313	314	315	316	317	318
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325	326	327	328	329	330
331	332	333	334	335	336
337	338	339	340	341	342
343	344	345	346	347	348
349	350	351	352	353	354
355	356	357	358	359	360
361	362	363	364	365	366
367	368	369	370	371	372
373	374	375	376	377	378
379	380	381	382	383	384
385	386	387	388	389	390
391	392	393	394	395	396
397	398	399	400	401	402
403	404	405	406	407	408
409	410	411	412	413	414
415	416	417	418	419	420
421	422	423	424	425	426
427	428	429	430	4	

Number of Logic Block				
-----------------------	--	--	--	--

Name of the School		Date	
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20
21	22	23	24
25	26	27	28
29	30	31	32
33	34	35	36
37	38	39	40
41	42	43	44
45	46	47	48
49	50	51	52
53	54	55	56
57	58	59	60
61	62	63	64
65	66	67	68
69	70	71	72
73	74	75	76
77	78	79	80
81	82	83	84
85	86	87	88
89	90	91	92
93	94	95	96
97	98	99	100

Number of Logic Block | _ | _ | _ |

[illegible]

Number of Logic Block | _ | _ | _ |

[illegible]

Number of Logic Block | _ | _ | _ |

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
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References: Logic Block Form Instructions in the Implementation/Hiway Gateway - 1 binder
Logic Block Data Entry in the Implementation/Hiway Gateway - 1 binder

Table 3-5 — Entries for Logic Block Comments Form

LINE	NAME	DATA ENTRY	REMARKS
		TEXT FILE EDITOR DISPLAY	
0.0	Pathname		Pathname entries are not required for logic-block configuration, but are recommended to facilitate data entry.
0.1	Device	NET (History Module) or \$Fn (floppy drive n)	
0.2	Volume	XXXX (user-defined in Network configuration)	
0.3	Filename	LBxxyyzz, where xx = Hiway Number (01-20). yy = Box Number (01-31). zz = Serial Number (00-99)	Configured in Network. Box Number of Multifunction Controller. User number for revision level of this filename.
1.0	Comments		
	No. of Logic Block	Enter the number of the logic block to which the comment applies. For each comment line, enter any character string (alphanumerics, symbols, and spaces).	Comments can apply to logic blocks before or after the comment, or to any group of logic blocks. This number is only a reference for locating comments in a Text Editor listing. The first two characters of each comment line, when entered into the system via the Text File Editor function must be hyphens, as preprinted on the comments form.

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Writer: **J. Danner**

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CITY _____ STATE _____ ZIP _____
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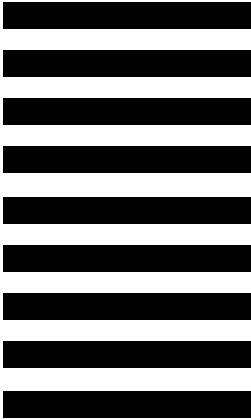
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